

Safe System of Work

Use of Excavators in Lifting Operations

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Overview

The following information provides guidance on the company's Standard Operation Procedures for works involving the use of Excavators for Lifting Operations on site.

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

Planning

Regulation 8 of LOLER requires lifting operations to be properly planned, appropriately supervised and carried out in a safe manner, and this includes lifting by earth moving machinery. The competent person (See LOLER ACOP and Guidance para. 220 –223) should first carry out a risk assessment, which will include ensuring that the earth moving machinery is suitable for the task, that adequate lifting accessories are available and the ground conditions are suitable. The lifting accessories, including quick hitches, should be suitable for the load and marked with their safe working load. The risk assessment will also address the need to segregate the lifting operation from other persons on site. Based on this risk assessment, a written method statement setting out the safe operating procedure to be followed should be prepared and communicated to all personnel involved in the lifting operation.

Special attachments for lifting (or other uses) should be compatible with the machine and coupling means (quick hitch) to which they are being attached. A competent person should verify that the combination is designed to take the load in this manner. Further guidance on planning lifting operations is given in BS 7121-1:2006.

Hazards Identified

- Adverse weather
- Crushing
- Cuts and abrasions
- Vehicle movements
- Falls from height
- Manual handling
- Slips & Trips

The use of an excavator or backhoe for lifting creates additional hazards for personnel in the vicinity. Under normal circumstances, personnel are kept away from the working area around the bucket of an excavator, as this is considered to be a hazardous area. Where the excavator is used for object handling however, the slinger has to be in the danger area in order to hook the load on to the hooking device. This puts the slinger at risk of being struck by the load, bucket or excavator arm if the excavator moves without warning.

One common incident of this type is where the excavator slews rapidly, for example when the operator catches the cuff of his coat on the left hand joystick, and the ground worker is struck by the bucket or dipper arm or crushed between the dipper arm and a fixed object. Injuries are usually severe in these

cases. Excavator operators and slingers should be made aware of these dangers and effective communications must always be maintained between slinger and operator. The excavator operator should always make use of the safety lever (safety armrest) whenever the slinger is in the danger area.

Traveling with a suspended load is particularly hazardous. The pre-lifting risk assessment should ensure that the hazards associated with this operation are fully considered, particularly the presence of personnel adjacent to the machine's travel path

Personal Protective Equipment (PPE) Requirements



Operations

Excavator Operator

- The excavator operator should be responsible for the correct operation of the excavator in accordance with the manufacturer's instructions and within the details of the lift plan.
- The excavator operator should respond only to the communications from the slinger/signaler, who should be clearly identified.

The excavator operator should:

- a. Have the necessary competence (skills, knowledge and experience) to carry out lifting operations.
- b. Be familiar with the excavator to be operated, check that it is in good condition and that it has sufficient capacity to carry out the lift safely.
- c. Ensure that they do not wear loose clothing, which could snag on the controls and lead to unintended movement.
- d. Ensure, before the lifting operation starts, that the bucket is removed from the machine if the lifting attachment (hook) is fitted to the quick hitch or dipper end.
- e. Ensure that lifting operations are only carried out with the excavator in lifting mode and the overload warning device or rated capacity indicator/limiter selected.
- f. Ensure that they have been briefed on and understand the lift plan.

- g. Identify the other members of the lifting team and ensure that they are clear of the excavator's arc before operating the machine.
- h. Check that the area where the excavator is to be positioned for the lifting operation is suitable for the task, the landing area is suitable to take the load, the area is segregated from the rest of the site and that only those personnel directly involved in the lift are within the working area.
- i. Ensure that the pre-use checks of the lifting accessories to be used have been carried out and that the lifting accessories have been correctly attached to the excavator's lifting attachment.
- j. Ensure that the excavator's control isolator (dead man) is selected when the lifting accessories and load are being attached to avoid unintended movement.
- k. Only follow signals from the designated slinger/signaler during the lifting operation, using the established and agreed pre-arranged system of communication.

NOTE: It is essential that the excavator operator responds immediately to an emergency stop signal from any person.

Slinger / Signaler

- The slinger/signaler should be properly trained in all aspects of slinging loads, signaling and working with excavators.

The slinger/signaler should be responsible for:

- a. Carrying out pre-use and post-use checks of lifting accessories.
- b. Attaching and detaching the load to and from the excavator load-lifting attachment.
- c. Using the correct lifting accessories and other equipment in accordance with the lift plan.
- d. Initiating and directing the safe movement of the excavator using an established and agreed pre-arranged system of communication. If there is more than one slinger/signaler, only one of them should have this responsibility at any one time, depending on their positions relative to the excavator.
- e. Guiding movements of the excavator during pick and carry lifting operations.
- f. Ensuring that they are readily identifiable as the designated slinger/signaler by the excavator operator.

NOTE: Movement of the excavator includes pick and carry duties.